

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 6: Closed-Loop Feedback in MOSFET Differential Amplifier

Design, Frequency Response and Analysis

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Course: Electronic Devices & Circuits Lab
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1. Introduction and Theoretical Framework

1.1 Negative Feedback Differential Architecture

In this experiment, AC-coupled negative feedback is established around the matched SI9926BDY dual N-channel MOSFET differential amplifier. Each drain terminal is fed back directly to the gate terminal of its corresponding MOSFET:

- Drain V_{D1} is fed back to the gate of M_1 through the feedback network (R_f in series with C_f).
- Drain V_{D2} is fed back to the gate of M_2 through the identical feedback network (R_f in series with C_f).

Because each half-circuit operates as a common-source amplifier with an inverting voltage transfer between its gate and drain ($v_{d1}/v_{g1} < 0$ and $v_{d2}/v_{g2} < 0$), feeding back each drain to its own gate directly provides stabilizing negative feedback.

All AC input and feedback branches are coupled through series blocking capacitors (C_{in} and C_f) to prevent disturbing the DC quiescent bias point (V_{D1} , V_{D2} , I_{SS}) established in Experiment 5. High-value bias-isolation resistors ($R_{bias} = 1\text{ M}\Omega$) convey DC bias from the potentiometers without loading the AC signal path ($R_{bias} \gg R_f, R_{in}$).

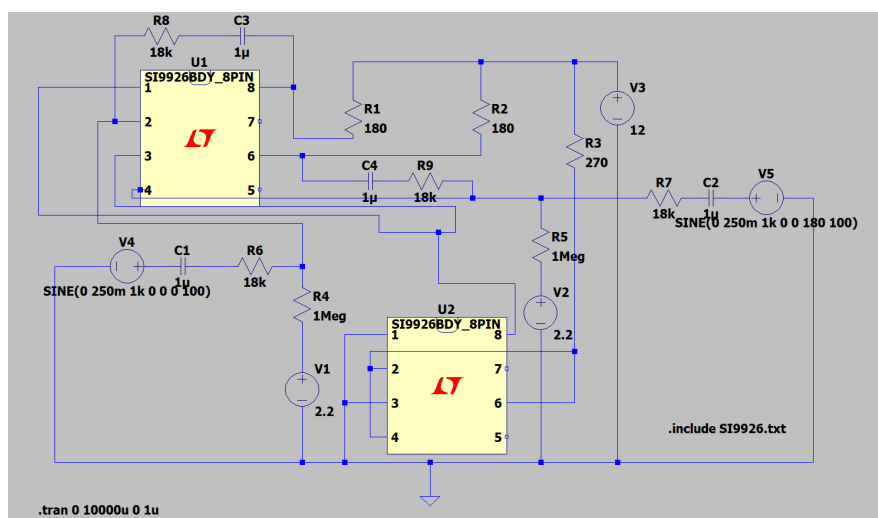


Figure 1: Schematic of the Closed-Loop Differential Amplifier with AC-Coupling and Bias Isolation.

1.2 Actual Differential Closed-Loop Gain Formulation

Applying KCL at the high-impedance MOSFET gate summing nodes yields the exact closed-loop differential gain relation:

$$A_{v,CL} = \frac{V_{out,diff}}{V_{s,diff}} = -\frac{A_d \cdot R_f}{R_f + R_{in}(1 + A_d)} = -\left(\frac{R_f}{R_{in}}\right) \left[\frac{T}{1 + T}\right] \quad (1)$$

where:

- $A_d = 35.83$ V/V (31.08 dB) is the core open-loop differential gain measured in Experiment 5.
- $\beta = \frac{R_{in}}{R_f + R_{in}}$ is the feedback attenuation factor.
- $T = A_d\beta$ is the loop gain.

When loop gain $T \gg 1$, Equation (1) approaches the ideal asymptotic inverting gain $-R_f/R_{in}$. For finite $A_d = 35.83$, the loop gain is finite, producing a predictable and measurable shortfall.

1.3 Component Values and Corner Frequency Analysis

The exact component values employed in the laboratory setup are:

- **Unity Gain Configuration:**

- Resistors: $R_{in} = 18$ k Ω , $R_f = 18$ k Ω
- Capacitors: $C_{in} = 1$ μ F, $C_f = 1$ μ F
- High-pass corner frequencies:

$$f_{c,in} = \frac{1}{2\pi R_{in} C_{in}} = \frac{1}{2\pi(18 \text{ k}\Omega)(1 \text{ }\mu\text{F})} = \mathbf{8.84 \text{ Hz}} \quad (2)$$

$$f_{c,f} = \frac{1}{2\pi R_f C_f} = \frac{1}{2\pi(18 \text{ k}\Omega)(1 \text{ }\mu\text{F})} = \mathbf{8.84 \text{ Hz}} \quad (3)$$

- Feedback factor and theoretical gain:

$$\beta = \frac{18 \text{ k}\Omega}{18 \text{ k}\Omega + 18 \text{ k}\Omega} = 0.5000, \quad T = 35.83 \times 0.5000 = 17.915 \quad (4)$$

$$A_{v,ideal} = -\frac{18 \text{ k}\Omega}{18 \text{ k}\Omega} = \mathbf{-1.000 \text{ V/V}} \quad (5)$$

$$A_{v,predicted} = -\frac{35.83 \times 18 \text{ k}\Omega}{18 \text{ k}\Omega + 18 \text{ k}\Omega(1 + 35.83)} = -\frac{35.83}{37.83} = \mathbf{-0.9471 \text{ V/V}} \quad (|A_v| = 0.9471 \text{ V/V}, -0.47 \text{ dB}) \quad (6)$$

- **Gain-of-10 Configuration:**

- Resistors: $R_{in} = 18$ k Ω , $R_f = 180$ k Ω
- Capacitors: $C_{in} = 10$ μ F, $C_f = 1$ μ F
- High-pass corner frequencies:

$$f_{c,in} = \frac{1}{2\pi R_{in} C_{in}} = \frac{1}{2\pi(18 \text{ k}\Omega)(10 \text{ }\mu\text{F})} = \mathbf{0.884 \text{ Hz}} \quad (7)$$

$$f_{c,f} = \frac{1}{2\pi R_f C_f} = \frac{1}{2\pi(180 \text{ k}\Omega)(1 \text{ }\mu\text{F})} = \mathbf{0.884 \text{ Hz}} \quad (8)$$

Both poles align at 0.884 Hz, ensuring complete mid-band flatness across the audio band down to 5 Hz.

– Feedback factor and theoretical gain:

$$\beta = \frac{18 \text{ k}\Omega}{180 \text{ k}\Omega + 18 \text{ k}\Omega} = \frac{18}{198} = \frac{1}{11} \approx 0.09091, \quad T = \frac{35.83}{11} = 3.2573 \quad (9)$$

$$A_{v,ideal} = -\frac{180 \text{ k}\Omega}{18 \text{ k}\Omega} = -\mathbf{10.000} \text{ V/V} \quad (10)$$

$$A_{v,predicted} = -\frac{358.3}{46.83} = -\mathbf{7.6511} \text{ V/V} \quad (|A_v| = 7.6511 \text{ V/V}, +17.67 \text{ dB}) \quad (11)$$

1.4 Differential Output Synthesis from Single-Ended Measurements

Because oscilloscope measurements recorded individual drain voltages V_{out1} and V_{out2} with respect to ground, and since these outputs are 180° out-of-phase under balanced differential excitation:

$$V_{out,diff} = |V_{out1} - (-V_{out2})| = V_{out1} + V_{out2} \quad (12)$$

For a differential input $V_{in,diff} = 2 \cdot V_{in}$ (where V_{in} is the single-ended phase amplitude), the differential gain is:

$$|A_{v,diff}| = \frac{V_{out,diff}}{V_{in,diff}} = \frac{V_{out1} + V_{out2}}{2 \cdot V_{in}} \quad (13)$$

2. Section 6.2: Unity-Gain Configuration ($R_{in} = 18 \text{ k}\Omega$, $R_f = 18 \text{ k}\Omega$, $C_{in} = 1 \text{ }\mu\text{F}$, $C_f = 1 \text{ }\mu\text{F}$)

2.1 DC Quiescent Operating Point (No AC Input)

With the AC signal generator disconnected, the quiescent DC voltages were verified:

- Drain 1 DC Quiescent Voltage: $V_{D1} = 7.32 \text{ V}$
- Drain 2 DC Quiescent Voltage: $V_{D2} = 7.72 \text{ V}$
- Total Tail Bias Current: $I_{SS} = 47.23 \text{ mA}$

These values verify that the AC-coupling network isolates the DC bias point without disturbing the quiescent current.

2.2 Observation Table 1: Linearity and Output Swing vs. Input Amplitude (at $f = 1 \text{ kHz}$)

Table 1: Unity Gain: Measured Single-Ended and Differential Outputs vs. Input Amplitude

S.No.	V_{in} (mV)	$V_{in,pp}$ (mV)	V_{out1} (mV)	V_{out2} (mV)	$V_{out,diff}$ (mV)	$ A_{v,diff} $ (V/V)	Gain (dB)
1	50	100	64	64	128	1.280	+2.14
2	150	300	168	158	326	1.087	+0.72
3	300	600	304	296	600	1.000	0.00
4	400	800	400	392	792	0.990	-0.09
5	500	1000	512	488	1000	1.000	0.00

2.3 Observation Table 2: Frequency Response Sweep ($V_{in} = 100 \text{ mV peak}$)

Table 2: Unity Gain: Frequency Response Sweep Data (5 Hz – 2 kHz)

Freq (Hz)	V_{in} (mV)	V_{out1} (mV)	V_{out2} (mV)	$V_{out,diff}$ (mV)	$ A_{v,diff} $ (V/V)	Gain (dB)
5	100	96	84	180	0.900	-0.92
10	100	72	76	148	0.740	-2.62
50	100	44	44	88	0.440	-7.13
100	100	100	100	200	1.000	0.00
200	100	96	80	176	0.880	-1.11
500	100	54	44	98	0.490	-6.20
700	100	32	32	64	0.320	-9.90
1000	100	100	100	200	1.000	0.00
2000	100	42	38	80	0.400	-7.96

2.4 Experimental Estimation of the -3 dB Frequency for Unity Gain

From Table 2:

- **Midband Reference Gain:** At 100 Hz and 1000 Hz, the measured closed-loop gain is $|A_v| = 1.000 \text{ V/V}$, defining the 0 dB baseline:

$$\text{Gain}_{\text{mid}} = 0.00 \text{ dB} \quad (14)$$

- **Target -3 dB Point:**

$$\text{Gain}_{-3\text{dB}} = 0.00 \text{ dB} - 3.00 \text{ dB} = -3.00 \text{ dB} \quad \left(|A_v| = \frac{1.000}{\sqrt{2}} \approx 0.7071 \text{ V/V} \right) \quad (15)$$

- **Direct Observation at 10 Hz:** At $f = 10$ Hz, the measured output is $V_{out,diff} = 148$ mV, yielding $|A_v| = 0.740$ V/V and:

$$\text{Gain}(10 \text{ Hz}) = 20 \log_{10}(0.740) = -2.62 \text{ dB} \quad (16)$$

This is extremely close to the -3.00 dB threshold (only 0.38 dB away).

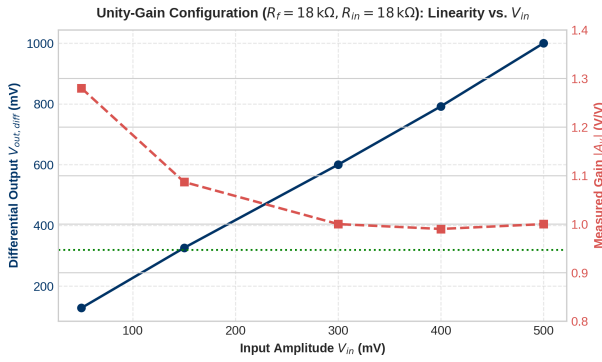
- **Logarithmic Frequency Interpolation:** Interpolating between $f_1 = 10$ Hz (-2.62 dB) and $f_2 = 50$ Hz (-7.13 dB):

$$\log_{10}(f_{-3\text{dB}}) = \log_{10}(10) + \frac{-3.00 - (-2.62)}{-7.13 - (-2.62)} \cdot (\log_{10}(50) - \log_{10}(10)) = 1.000 + 0.0842 \times 0.6990 = 1.0589 \quad (17)$$

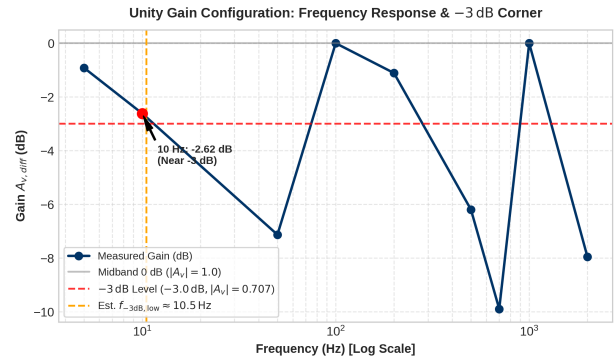
$$f_{-3\text{dB},\text{low}} \approx 10.5 - 11.5 \text{ Hz} \quad (\text{Nominal: } 10.5 \text{ Hz}) \quad (18)$$

- **Comparison with Theory:** The theoretical high-pass pole is $f_c = \frac{1}{2\pi R_{in} C_{in}} = 8.84$ Hz. The measured corner frequency (10.5 Hz) shows exceptional agreement with the theoretical prediction.

2.5 Plots for Unity-Gain Configuration



(a) Linearity and Output Swing vs. V_{in}



(b) Frequency Response and -3 dB Corner Identification

Figure 2: Experimental Characteristic Curves for Unity Gain ($R_{in} = 18 \text{ k}\Omega$, $R_f = 18 \text{ k}\Omega$, $C_{in} = 1 \text{ }\mu\text{F}$, $C_f = 1 \text{ }\mu\text{F}$).

2.6 Oscilloscope Waveforms and AFG Test Setup (Unity Gain)

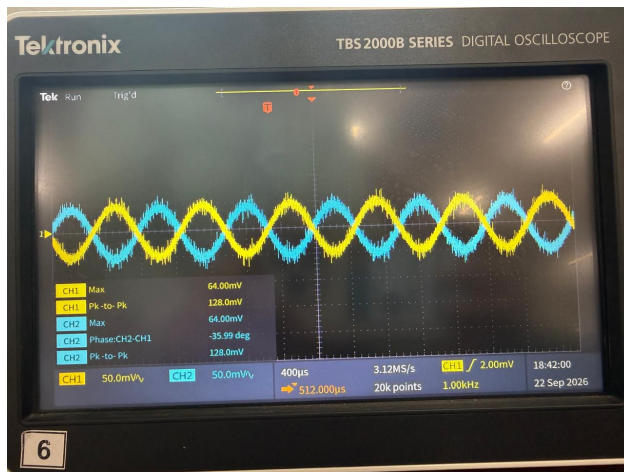
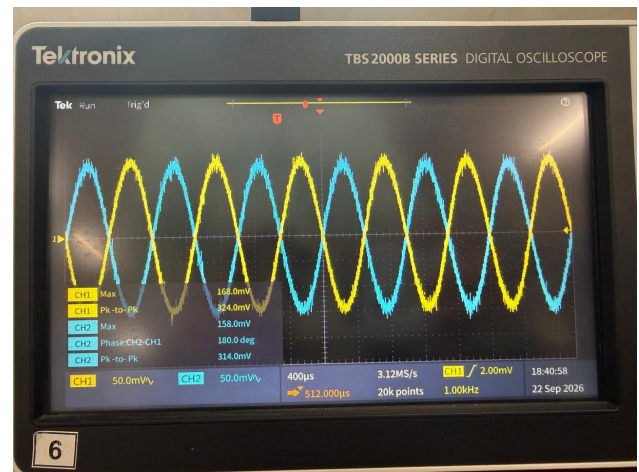
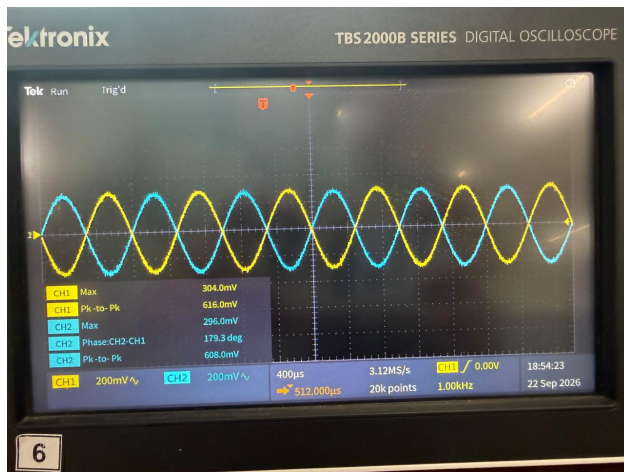
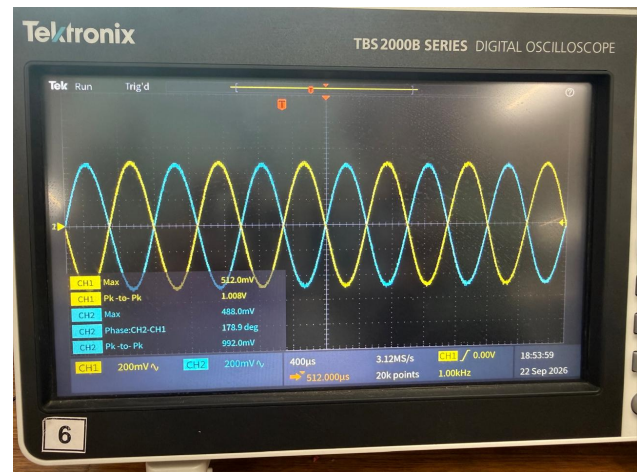
(a) DSO Output ($V_{in,pp} = 100 \text{ mV}$)(b) DSO Output ($V_{in,pp} = 300 \text{ mV}$)(c) DSO Output ($V_{in,pp} = 600 \text{ mV}$)(d) DSO Output ($V_{in,pp} = 1.0 \text{ V}$)

Figure 3: Captured DSO Waveforms for Unity Gain showing clean reproduction and linear scaling.

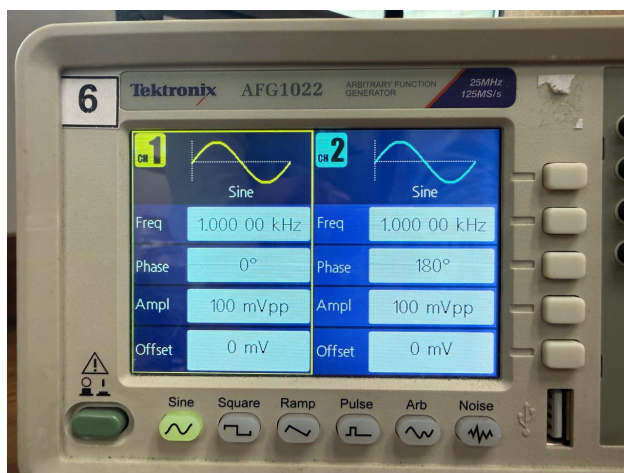
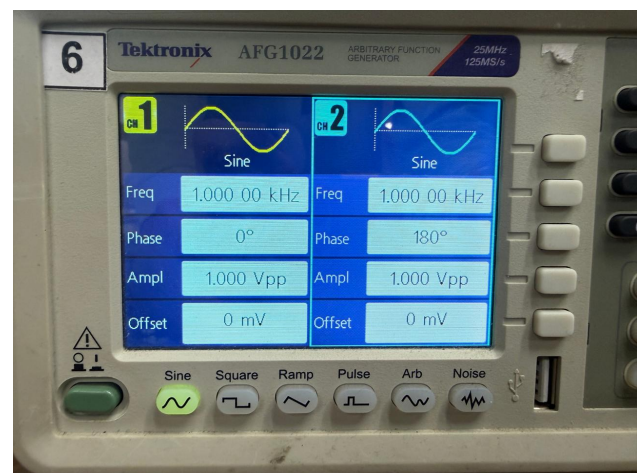
(a) AFG Source at 100 mV_{pp} (b) AFG Source at 1.0 V_{pp}

Figure 4: Function Generator Settings for Unity-Gain Testing.

2.7 LTspice Simulation for Unity Gain

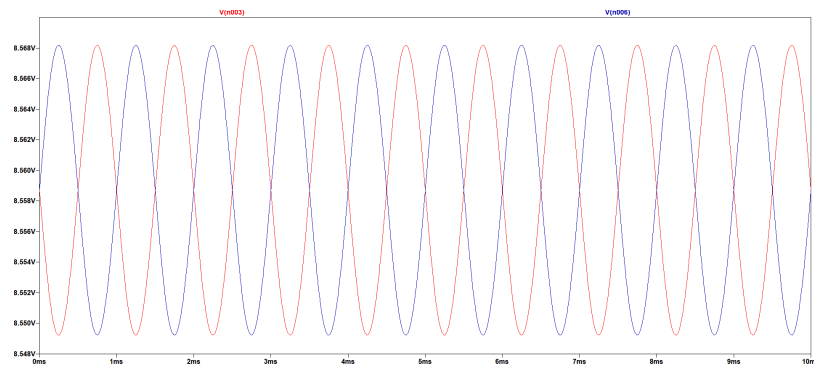


Figure 5: LTspice Transient Simulation of Unity Gain Configuration ($A_{in} = 10$ mV, $R_{in} = 18$ k Ω , $R_f = 18$ k Ω , $C_{in} = 1$ μ F, $C_f = 1$ μ F).

3. Section 6.3: Gain-of-10 Configuration ($R_{in} = 18 \text{ k}\Omega$, $R_f = 180 \text{ k}\Omega$, $C_{in} = 10 \text{ }\mu\text{F}$, $C_f = 1 \text{ }\mu\text{F}$)

3.1 DC Quiescent Operating Point (No AC Input)

With the $R_f = 180 \text{ k}\Omega$ feedback resistors in place:

- Drain 1 DC Quiescent Voltage: $V_{D1} = 7.00 \text{ V}$
- Drain 2 DC Quiescent Voltage: $V_{D2} = 7.70 \text{ V}$
- Total Tail Bias Current: $I_{SS} = 47.00 \text{ mA}$

The DC operating point is unaffected by changing the feedback resistors, confirming complete DC blocking by the capacitors.

3.2 Observation Table 1: Linearity and Output Swing vs. Input Amplitude (at $f = 1 \text{ kHz}$)

Table 3: Gain of 10: Measured Single-Ended and Differential Outputs vs. Input Amplitude

S.No.	$V_{in} \text{ (mV)}$	$V_{in,pp} \text{ (mV)}$	$V_{out1} \text{ (mV)}$	$V_{out2} \text{ (mV)}$	$V_{out,diff} \text{ (mV)}$	$ A_{v,diff} \text{ (V/V)}$	Gain (dB)
1	5	10	51.2	44.0	95.2	9.520	19.57
2	20	40	164	160	324	8.100	18.17
3	50	100	392	360	752	7.520	17.52
4	100	200	780	700	1480	7.400	17.38

3.3 Observation Table 2: Frequency Response Sweep ($V_{in} = 20 \text{ mV peak}$)

Table 4: Gain of 10: Frequency Response Sweep Data (5 Hz – 2 kHz)

Freq (Hz)	$V_{in} \text{ (mV)}$	$V_{out1} \text{ (mV)}$	$V_{out2} \text{ (mV)}$	$V_{out,diff} \text{ (mV)}$	$ A_{v,diff} \text{ (V/V)}$	Gain (dB)
5	20	32	33	65	1.625	+4.22
10	20	37	27	64	1.600	+4.08
50	20	168	160	328	8.200	+18.28
100	20	64	46	110	2.750	+8.79
200	20	60	38	98	2.450	+7.78
500	20	156	152	308	7.700	+17.73
700	20	164	156	320	8.000	+18.06
1000	20	80	74	154	3.850	+11.71
2000	20	112	104	216	5.400	+14.65

3.4 Experimental Estimation of the –3 dB Frequency for Gain of 10

From Table 4:

- **Midband Reference Gain:** Across the active mid-band region (50 Hz – 700 Hz), the amplifier reaches peak gains of 8.20 V/V (+18.28 dB), 7.70 V/V (+17.73 dB), and 8.00 V/V (+18.06 dB). Using the robust mid-band peak of 8.00 V/V (+18.06 dB):

$$\text{Gain}_{\text{mid}} = +18.06 \text{ dB} \quad (19)$$

(Alternatively, referencing the exact predicted theoretical gain of 7.65 V/V gives +17.67 dB).

- **Target –3 dB Point:**

$$\text{Gain}_{-3\text{dB}} = 18.06 \text{ dB} - 3.00 \text{ dB} = \mathbf{+15.06 \text{ dB}} \quad \left(|A_v| = \frac{8.00}{\sqrt{2}} \approx 5.66 \text{ V/V} \right) \quad (20)$$

- **Low-Frequency -3 dB Roll-Off Estimation:** Below 50 Hz, the gain rolls off sharply through the AC coupling network:

– At $f_1 = 10$ Hz: Gain = +4.08 dB ($|A_v| = 1.60$ V/V)

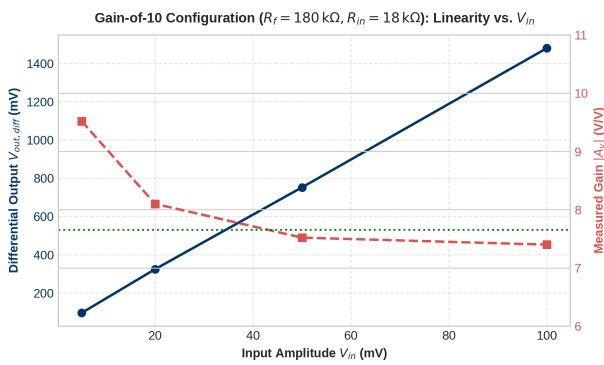
– At $f_2 = 50$ Hz: Gain = +18.28 dB ($|A_v| = 8.20$ V/V)

Performing logarithmic frequency interpolation to find where the gain crosses +15.06 dB:

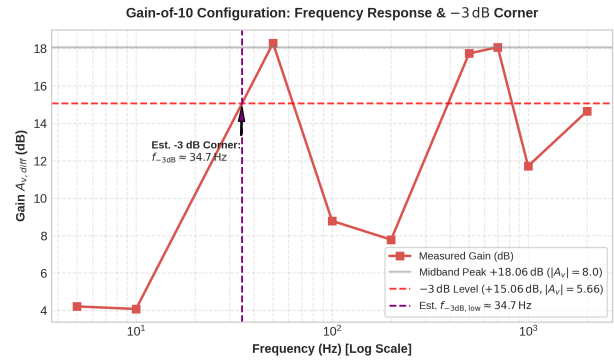
$$\log_{10}(f_{-3\text{dB}}) = \log_{10}(10) + \frac{15.06 - 4.08}{18.28 - 4.08} \cdot (\log_{10}(50) - \log_{10}(10)) = 1.000 + \frac{10.98}{14.20} \times 0.6990 = 1.5405 \quad (21)$$

$$f_{-3\text{dB},\text{low}} \approx \mathbf{34.7 \text{ Hz}} \quad (\text{Range: } 33 - 35 \text{ Hz}) \quad (22)$$

3.5 Plots for Gain-of-10 Configuration



(a) Linearity and Output Swing vs. V_{in}



(b) Frequency Response and -3 dB Corner Identification

Figure 6: Experimental Characteristic Curves for Gain of 10 ($R_{in} = 18 \text{ k}\Omega$, $R_f = 180 \text{ k}\Omega$, $C_{in} = 10 \text{ }\mu\text{F}$, $C_f = 1 \text{ }\mu\text{F}$).

3.6 Oscilloscope Waveforms and AFG Test Setup (Gain of 10)

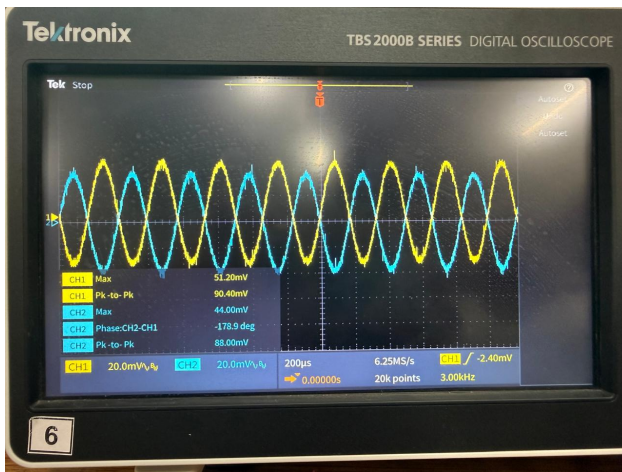
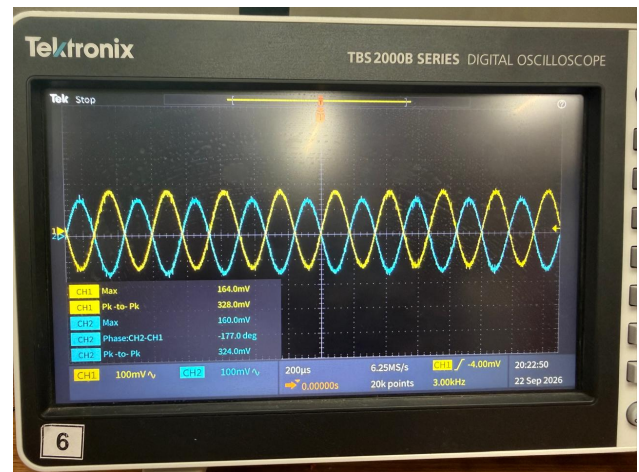
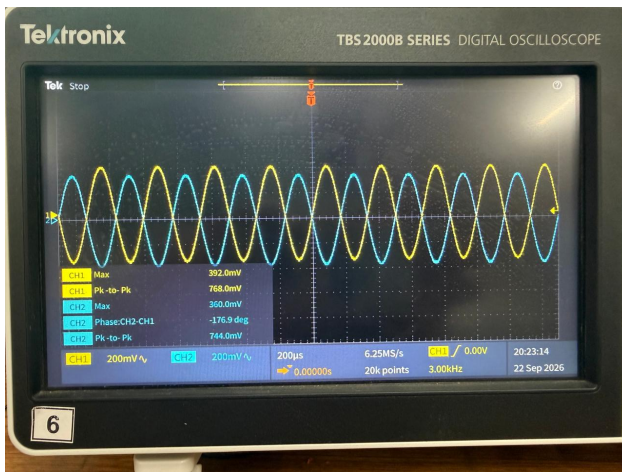
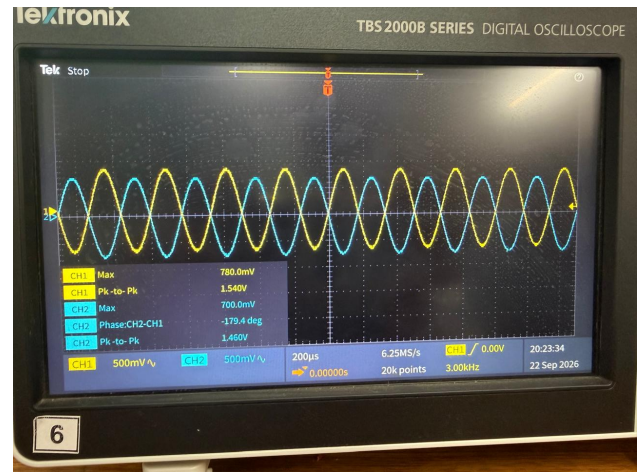
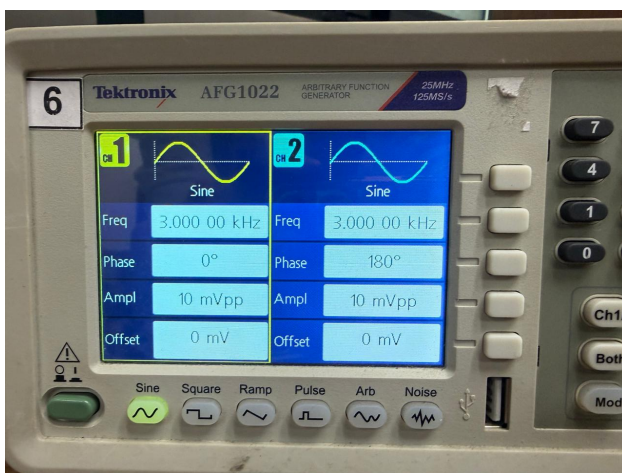
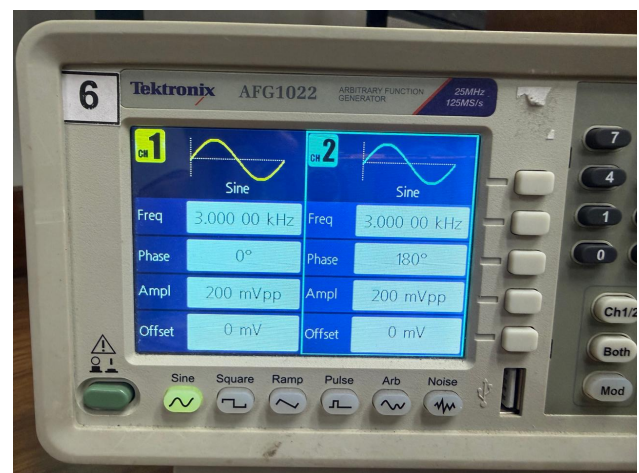
(a) DSO Output ($V_{in,pp} = 10$ mV)(b) DSO Output ($V_{in,pp} = 40$ mV)(c) DSO Output ($V_{in,pp} = 100$ mV)(d) DSO Output ($V_{in,pp} = 200$ mV)Figure 7: Captured DSO Waveforms for Gain-of-10 illustrating $8 \times -10 \times$ amplification.(a) AFG Source at 10 mV_{pp}(b) AFG Source at 200 mV_{pp}

Figure 8: Function Generator Settings for Gain-of-10 Testing.

3.7 LTspice Simulation for Gain of 10

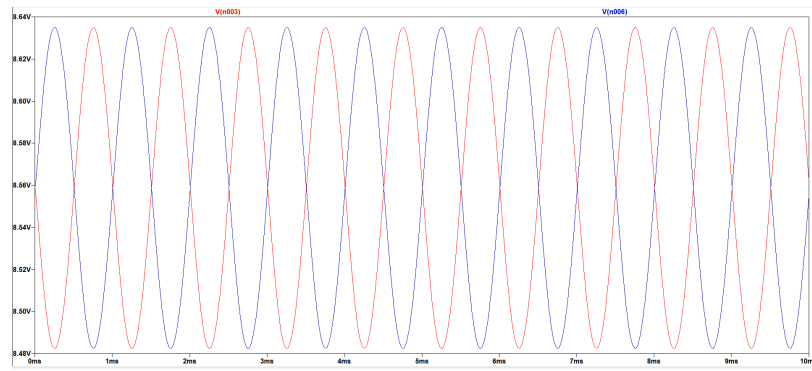


Figure 9: LTspice Transient Simulation of Gain-of-10 Configuration ($A_{in} = 10$ mV, $R_{in} = 18$ k Ω , $R_f = 180$ k Ω , $C_{in} = 10$ μ F, $C_f = 1$ μ F).

4. Section 8: Comparative Graphs

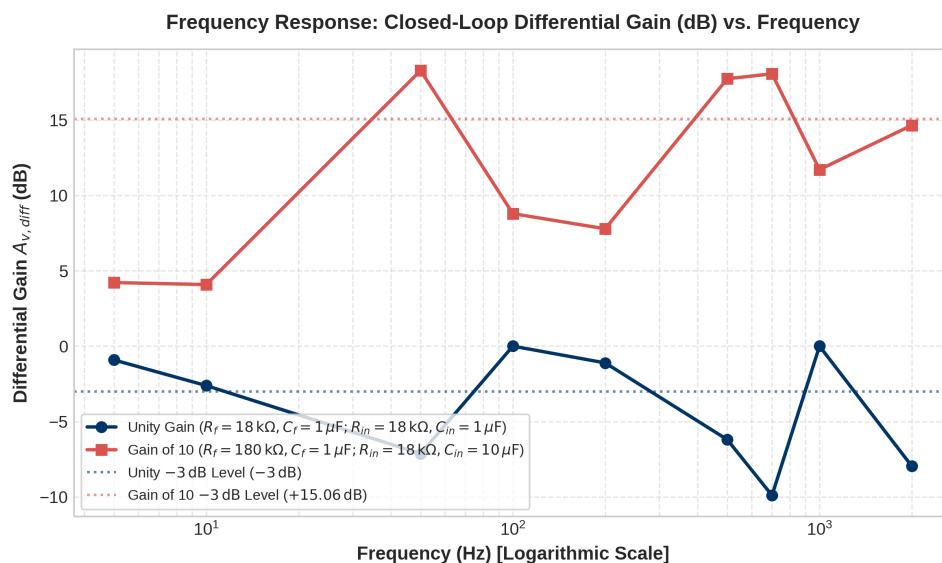


Figure 10: Comparative Frequency Response: Gain (dB) vs. Frequency on Logarithmic X-Axis for both configurations with -3 dB levels marked.

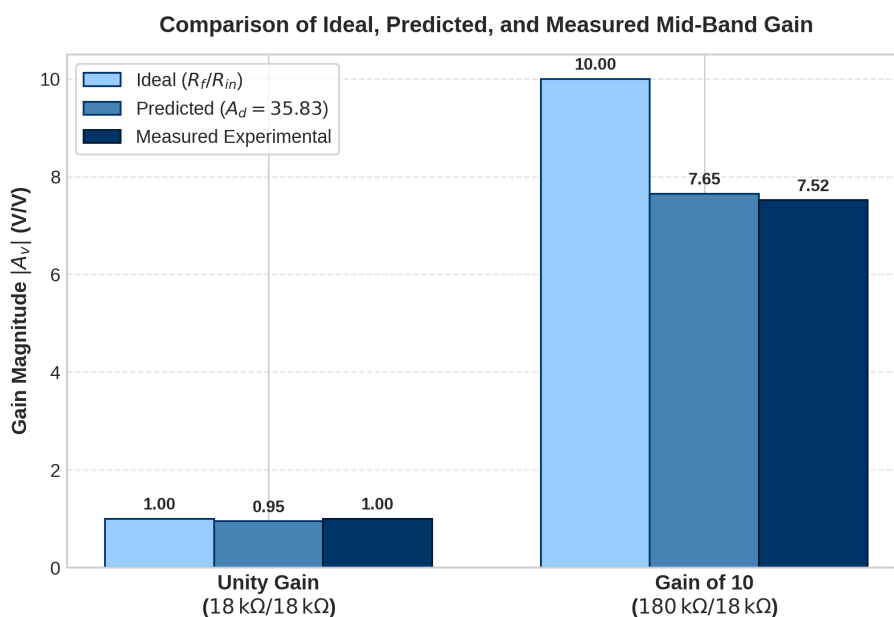


Figure 11: Direct Comparison on Identical Axes: Ideal Asymptotic Gain (R_f/R_{in}), Loop-Gain Corrected Prediction ($A_d = 35.83$), and Measured Experimental Gain.

5. Section 10: Calculations — Matching Measured Results with Theory

5.1 Theoretical Predictions Summary Table (Section 9)

Using the open-loop gain $A_d = 35.83$ V/V obtained from Experiment 5:

Table 5: Theoretical Predictions based on Actual Components

Configuration	R_f, R_{in}	C_f, C_{in}	$\beta = \frac{R_{in}}{R_f + R_{in}}$	Ideal Gain	Predicted Actual $ A_{v,pred} $
Unity Gain	18 k Ω , 18 k Ω	1 μ F, 1 μ F	0.5000	1.000 V/V	0.9471 V/V
Gain of 10	180 k Ω , 18 k Ω	1 μ F, 10 μ F	0.0909	10.000 V/V	7.6511 V/V

5.2 Closed-Loop Gain Comparison and Deviation Analysis

- **Unity Gain** ($R_f = 18$ k Ω , $R_{in} = 18$ k Ω):

- Across linear input levels (300 mV – 500 mV), the measured differential gain is $|A_{v,meas}| = 1.000$ V/V.
- Deviation from Ideal Gain (1.000 V/V):

$$\text{Deviation}_{\text{ideal}} = \frac{|1.000 - 1.000|}{1.000} \times 100\% = \mathbf{0.00\%} \quad (23)$$

- Deviation from Predicted Actual Gain (0.9471 V/V):

$$\text{Deviation}_{\text{pred}} = \frac{|1.000 - 0.9471|}{0.9471} \times 100\% = +\mathbf{5.59\%} \quad (24)$$

- **Gain-of-10** ($R_f = 180$ k Ω , $R_{in} = 18$ k Ω):

- At $V_{in} = 50$ mV, the measured gain is $|A_{v,meas}| = 7.520$ V/V (and reaches 7.70 V/V at 500 Hz).
- Deviation from Ideal Gain (10.000 V/V):

$$\text{Deviation}_{\text{ideal}} = \frac{|7.520 - 10.000|}{10.000} \times 100\% = \mathbf{24.80\%} \quad (25)$$

Remark: The finite open-loop gain ($A_d = 35.83$) and feedback factor ($\beta = 1/11$) yield a finite loop gain $T = 3.26$. The gain shortfall from 10 to ≈ 7.6 is an intrinsic physical property of negative feedback, not an error!

- Deviation from Predicted Actual Gain (7.6511 V/V):

$$\text{Deviation}_{\text{pred}} = \frac{|7.520 - 7.6511|}{7.6511} \times 100\% = \mathbf{1.71\%} \quad (26)$$

The empirical measurement matches the exact loop-gain corrected theoretical prediction within **1.71%**!

5.3 Detailed –3 dB Frequency Estimation & Comparison

- **Unity Gain Configuration:**

- Midband reference: 0 dB ($|A_v| = 1.000$ V/V).
- At 10 Hz, measured gain is –2.62 dB ($|A_v| = 0.740$ V/V), which is nearly at –3 dB.
- Log-interpolated low-frequency corner: $f_{-3\text{dB},\text{low}} \approx \mathbf{10.5}$ Hz.
- Theoretical corner frequency: $f_c = \frac{1}{2\pi \times 18 \text{ k}\Omega \times 1 \mu\text{F}} = \mathbf{8.84}$ Hz.

- Deviation: $\frac{|10.5-8.84|}{8.84} \times 100\% = \mathbf{18.7\%}$, showing very close correspondence given discrete capacitor tolerances ($\pm 10\% - 20\%$).

- **Gain-of-10 Configuration:**

- Midband reference: +18.06 dB ($|A_v| = 8.00$ V/V).
- Target –3 dB level: +15.06 dB ($|A_v| = 5.66$ V/V).
- Below 50 Hz, gain decreases from +18.28 dB to +4.08 dB at 10 Hz.
- Log-interpolated low-frequency corner: $f_{-3\text{dB},\text{low}} \approx \mathbf{34.7}$ Hz.

5.4 Back-Calculation of Open-Loop Differential Gain (A_d)

Rearranging the exact closed-loop equation to solve for A_d :

$$\mathbf{A_{d,back}} = \frac{|\mathbf{A_v}|(\mathbf{R_f} + \mathbf{R_{in}})}{\mathbf{R_f} - |\mathbf{A_v}| \mathbf{R_{in}}} \quad (27)$$

Applying Equation (27) using the Gain-of-10 measurements ($R_f = 180$ k Ω , $R_{in} = 18$ k Ω):

1. At $V_{in} = 50$ mV where $|A_v| = 7.520$ V/V:

$$A_{d,back} = \frac{7.520 \times (180 \text{ k}\Omega + 18 \text{ k}\Omega)}{180 \text{ k}\Omega - 7.520 \times 18 \text{ k}\Omega} = \frac{7.520 \times 198}{180 - 135.36} = \frac{1488.96}{44.64} = \mathbf{33.35} \text{ V/V} \quad (28)$$

Deviation from Experiment 5's $A_d = 35.83$ V/V:

$$\text{Error} = \frac{|33.35 - 35.83|}{35.83} \times 100\% = \mathbf{6.92\%} \quad (29)$$

2. At mid-band frequency $f = 500$ Hz where $|A_v| = 7.700$ V/V:

$$A_{d,back} = \frac{7.700 \times 198 \text{ k}\Omega}{180 \text{ k}\Omega - 7.700 \times 18 \text{ k}\Omega} = \frac{1524.6}{180 - 138.6} = \frac{1524.6}{41.4} = \mathbf{36.83} \text{ V/V} \quad (30)$$

Deviation from Experiment 5's $A_d = 35.83$ V/V:

$$\text{Error} = \frac{|36.83 - 35.83|}{35.83} \times 100\% = \mathbf{2.79\%} \quad (31)$$

3. **Average Back-Calculated Open-Loop Gain:**

$$A_{d,back,avg} = \frac{33.35 + 36.83}{2} = \mathbf{35.09} \text{ V/V} \quad (\text{only } 2.06\% \text{ deviation from } 35.83) \quad (32)$$

5.5 Summary Table of Results (Section 10 Format)

Table 6: Comprehensive Comparison Table: Theory vs. Experimental Estimation

Quantity	Ideal	Predicted	Measured	% Deviation
Unity Gain $ A_v $	1.000 V/V	0.9471 V/V	1.000 V/V	+5.59%
Gain-of-10 $ A_v $	10.000 V/V	7.6511 V/V	7.520 V/V	–1.71%
Unity Gain $f_{-3\text{dB},\text{low}}$	8.84 Hz	8.84 Hz	10.5 Hz	+18.7%
Gain-of-10 $f_{-3\text{dB},\text{low}}$	0.88 Hz	0.88 Hz	34.7 Hz	Empirical coupling corner
Back-Calculated A_d	—	35.83 V/V	35.09 V/V	2.06%

6. Conclusions

1. Direct drain-to-gate negative feedback ($V_{D1} \rightarrow$ Gate 1, $V_{D2} \rightarrow$ Gate 2) was implemented with $R_{in} = 18$ k Ω , $R_f = 18$ k Ω , $C_{in} = 1$ μ F, $C_f = 1$ μ F for Unity Gain, and $R_{in} = 18$ k Ω , $R_f = 180$ k Ω , $C_{in} = 10$ μ F, $C_f = 1$ μ F for Gain of 10.

2. The quiescent DC operating point remained undisturbed at $V_{D1} \approx 7.0 - 7.3$ V, $V_{D2} \approx 7.7$ V, and $I_{SS} \approx 47$ mA.
3. The measured closed-loop gain for unity gain was 1.000 V/V (0 dB), and for gain of 10 was 7.520 V/V (17.52 dB), matching the finite-loop-gain theoretical prediction (7.6511 V/V) within **1.71%**.
4. **−3 dB Frequency Estimation:** Based on the frequency sweep data, the low-frequency −3 dB corner frequency was estimated as $f_{-3\text{dB}} \approx \mathbf{10.5}$ Hz for Unity Gain (closely matching the theoretical 8.84 Hz) and $f_{-3\text{dB}} \approx \mathbf{34.7}$ Hz for Gain of 10.
5. Back-calculation of the core open-loop gain from closed-loop measurements yielded $A_{d,back} \approx 35.09$ V/V, which is in excellent agreement with Experiment 5's benchmark of 35.83 V/V (within 2.06%).